

Chapter 2: Reference Model

Protocols and Standards

Protocol is defined as a set of rules that governs any task. In computer network, it is a set of rules that governs the communication between two ends of communication. Different protocols exist at different levels during communication. eg. In OSI model, there are 7 layers and each layer has certain protocols to perform one or more task. Some network protocols are HTTP (Hyper Text Transfer Protocol), FTP (File Transfer protocol), etc

Standards are guidelines that are common to all the stakeholders. It explains how a particular protocol should operate.

De facto standards are that which are not approved by an organized body but are adopted as standards through widespread use. eg., MS Office

De jure are standards that have been legislated by an officially recognized body.

Some standards organizations are ISO (International Standards Organization), ANSI (American National Standards Institute), IEEE (Institute of Electrical and Electronics Engineers), EIA (Electronic Industries Alliance), etc

Interfaces and Services

In layered network architecture, there are a set of layers and protocols. The layers are organized as a stack, with each one built upon the one below it. A layered architecture with four layers is shown in following figure.

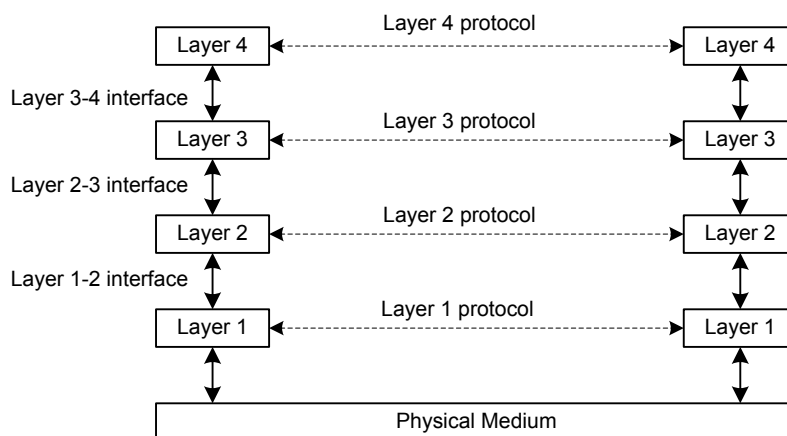


Fig: layer, protocol and interfaces

Each layer defines a family of functions distinct from those of the other layers. By defining and localizing functionality in this fashion, the designers created an architecture that is both comprehensive and flexible.

Interfaces

There is a well defined interface between each pair of adjacent layers that make it possible for the data to pass down through the layers at the sending host and back up through the same set of layers at the receiving host. Each interface defines the information, operation and services a layer must provide for the layer above it. Well-defined interface and layer functions provide greater modularity to a network. As long as a layer provides the expected services to the layers above it a completely different one can replace the implementation.

Services

Within a host, each layer calls upon the services of the layer just below it. For example, Layer 3 uses the service provided by layer 2 and provides its own services for layer 4. Between machines, layer x on one communicates with the corresponding layer x on another machine. This communication is governed by agreed rules of services called protocols. In reality no data are directly transferred from layer x on one machine to the same layer on another machine. Instead, each layer passed data and control information to the layer immediately below it, until the lowest layer is reached. Below layer 1 is the physical medium through which actual communication occur.

Importance of layered architecture

1. Layering reduces the design complexity.

It enables peer process abstraction which makes the unmanageable task of designing the complete network possible to be broken into several smaller manageable design problems, that is, the design of individual layers.

2. Provision of localized services strengthens modularity.

Each layer offers certain services to the higher layers and thus shields those layers from the details of how the offered services are actually implemented. Interfaces on all machines in a network need not be the same, if each machine correctly uses all the protocols.

3. They allow complete interoperability between incompatible systems

By defining the complete layered network model with all the interface details, the services and the protocols it is possible to ensure interoperability among various systems. There can be a seamless operation without considering the operating system or the computer hardware. For example, the OSI model has been developed with this aim.

4. Hardware and software vendor independence.

As long as correct layer functionality is ensured a network can be set up using products from different vendors and still the operation is guaranteed. For example, repeaters, hubs in the physical layer and web browser in the application layer.

OSI MODEL

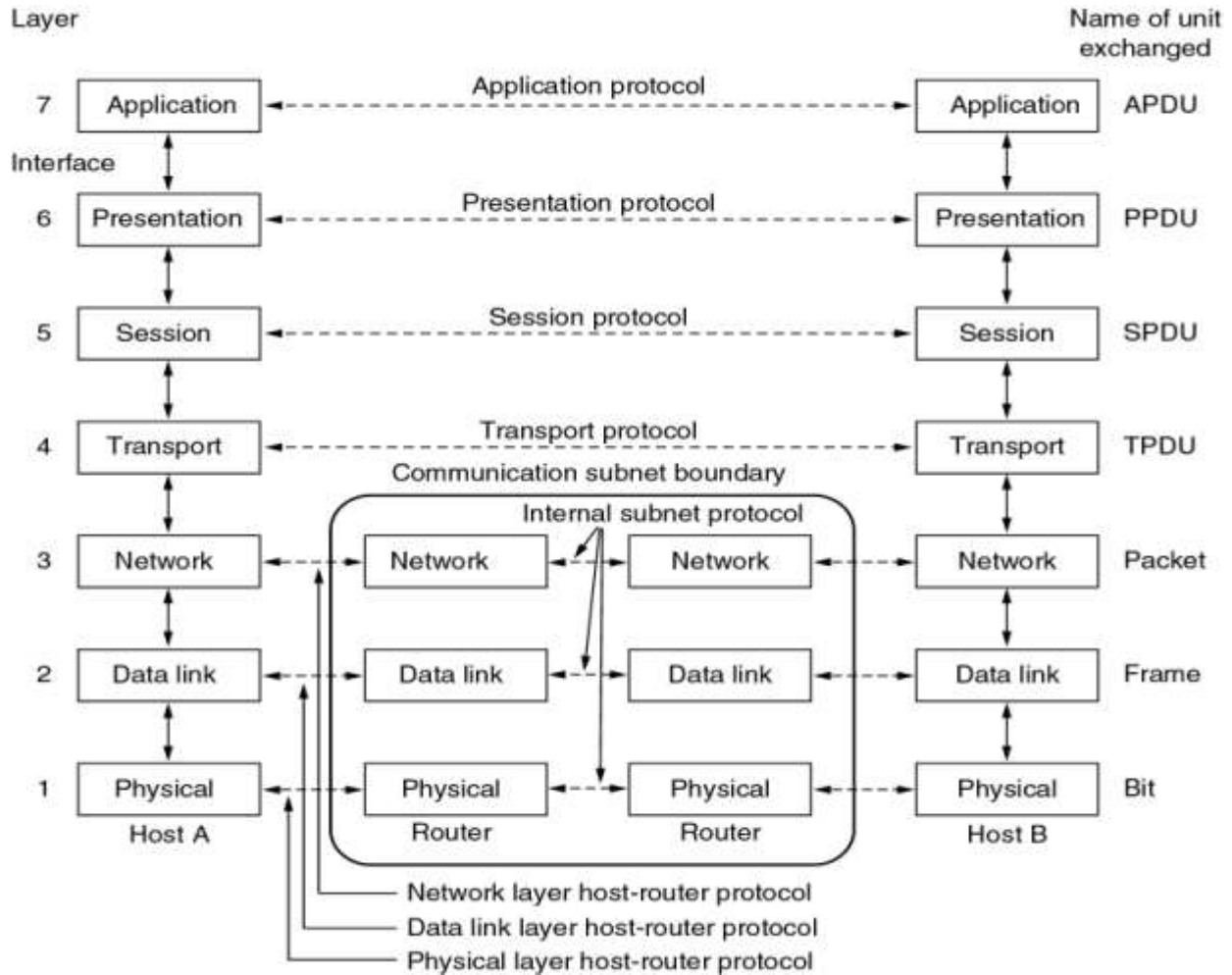


Figure: OSI model

The Physical Layer

The physical layer is concerned with transmitting raw bits over a communication channel. The design issues largely deal with mechanical, electrical, and timing interfaces, as well as the physical transmission medium.

The Data Link Layer

The main task of the data link layer is to transform a raw transmission facility into a line that appears free of undetected transmission errors. It break up the input data into **data frames** (typically a few hundred or a few thousand bytes) and transmit the frames sequentially. Another issue that arises in the data link layer (and most of the higher layers as well) is how to keep a fast transmitter from drowning a slow receiver in data (i.e., flow control).

The Network Layer

The network layer controls the operation of the subnet. A key design issue is determining how packets are routed from source to destination. If too many packets are present in the subnet at the same time, they will get in one another's way, forming bottlenecks. Handling congestion is also a responsibility of the network layer, in conjunction with higher layers.

The Transport Layer

The basic function of the transport layer is to accept data from above it, split it up into smaller units if need be, pass these to the network layer, and ensure that the pieces all arrive correctly at the other end. The transport layer also determines what type of service to provide to the session layer, and, ultimately, to the users of the network.

The Session Layer

The session layer allows users on different machines to establish sessions between them. Sessions offer various services, including **dialog control** (keeping track of whose turn it is to transmit), **token management** (preventing two parties from attempting the same critical operation simultaneously), and **synchronization** (check-pointing long transmissions to allow them to pick up from where they left off in the event of a crash and subsequent recovery).

The Presentation Layer

Unlike the lower layers, which are mostly concerned with moving bits around, the presentation layer is concerned with the syntax and semantics of the information transmitted.

The Application Layer

The application layer contains a variety of protocols that are commonly needed by users. One widely used application protocol is **HTTP (HyperText Transfer Protocol)**, which is the basis for the World Wide Web.

TCP/IP MODEL

There are 4 layers in the TCP/IP model

Layer 4: Application

Layer 3: Transport

Layer 2: Internet

Layer 1: Network access (Host to Network)

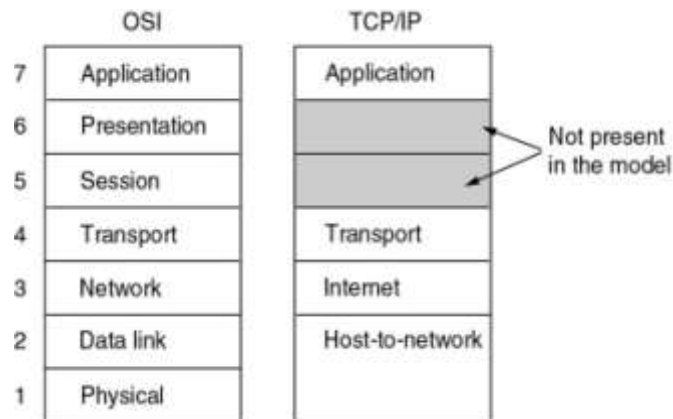


Fig: OSI and TCP/IP model

The network access layer (Host to Network)

Concerned with all of the issues that an IP packet requires to actually make the physical link. It includes all the details in the OSI physical and data link layers.

- Electrical, mechanical, procedural and functional specifications
- Data rate, Distances, Physical connector
- Frames
- Synchronization, flow control, error control

The Internet Layer

It is concerned with Packet Addressing. It sends source packets from any network and have them arrive at the destination independent of the path and networks they took to get there. Packet may arrive in different order to destination. It is job of higher layer to arrange them. It is also important to avoid congestion while packet routing.

The Transport Layer

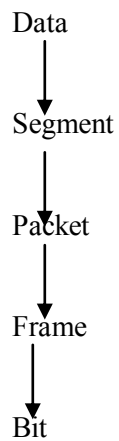
The transport layer deals with the quality-of-service issues of reliability, flow control, and error correction. Two end-to-end transport protocols have been defined here. The first one, **TCP (Transmission Control Protocol)**, is a reliable connection-oriented protocol that allows a byte stream originating on one machine to be delivered without error on any other machine in the internet. The second protocol in this layer, **UDP (User Datagram Protocol)**, is an unreliable, connectionless protocol for applications that do not want TCP's sequencing or flow control and wish to provide their own.

The Application Layer

The TCP/IP model does not have session or presentation layers. Instead, applications simply include any session and presentation functions that they require. On top of the transport layer is the application layer. It contains all the higher-level protocols. The early ones included virtual terminal (TELNET), file transfer (FTP), and electronic mail (SMTP).

How data encapsulation and de-encapsulation occurs in OSI model?

Data Encapsulation



Data De-encapsulation

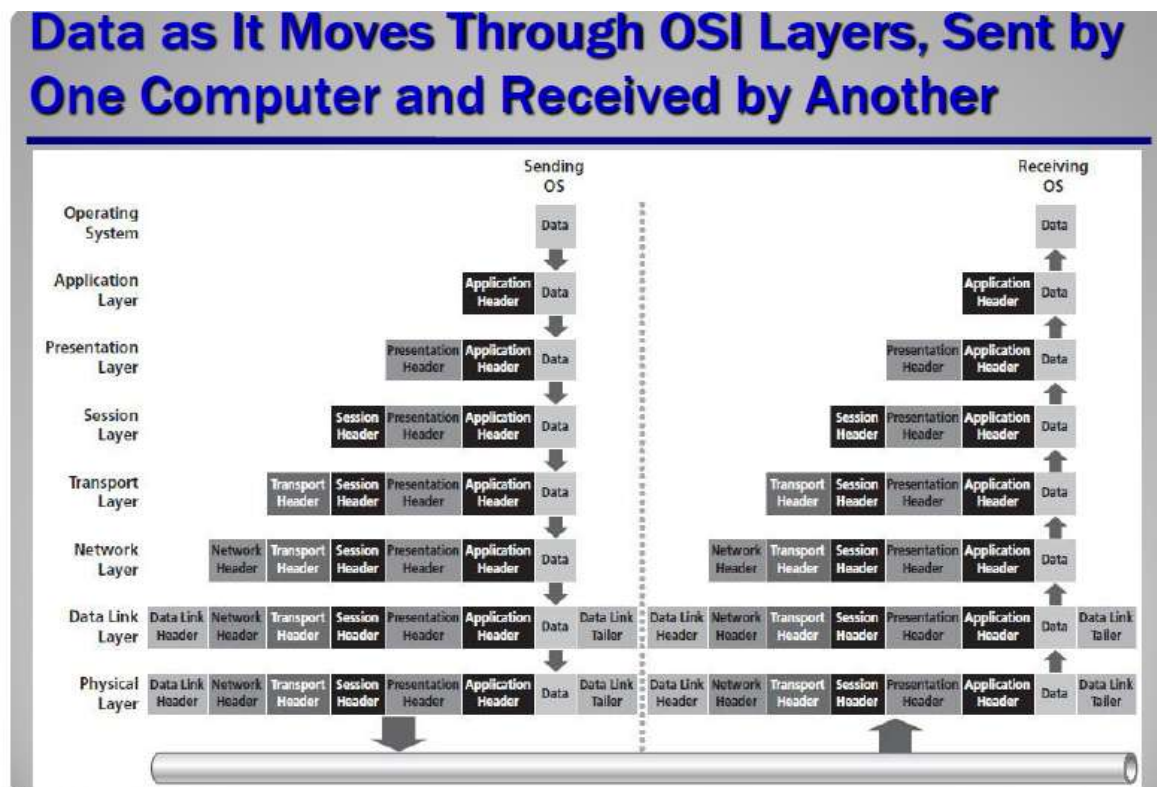
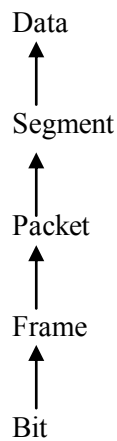


Fig: Data encapsulation and De-encapsulation

Comparison of OSI and TCP/IP

S.N	OSI	TCP/IP
1	It has 7 layers	It has 4 layers
2	OSI model has separate session layer and presentation layer	TCP/IP does not have separate session or presentation layer
3	Transport layer is connection oriented	Transport layer is both connection oriented and connectionless
4	Network layer is both connection oriented and connectionless	Network layer is connectionless
5	OSI is a reference model around which network are built. Generally used as guidance tool	TCP/IP is implementation model around which internet is based.
6	OSI model has a problem of fitting the protocol into the model	TCP/IP model does not fit any protocol
7	Protocols are hidden in OSI model and are easily replaced as the technology changes	In TCP/Ip model, replacing protocol is not easy
8	Concepts central to the OSI model are Services, Interfaces and Protocols	TCP/IP did not originally distinguish between services, interface and protocols.

Networking Hardware: NIC, Hub, Repeaters, Switches, Bridge, Router

NIC (Network Interface Card)

It is a circuit board or card that allows computers to communicate over a network via cables or wirelessly. It is also called network adaptor or network card. It enables clients, servers, printers and other devices to transmit and receive data over the network.

Hub

It is also called An Ethernet hub, active hub, network hub, repeater hub, multiport repeater or hub. It has multiple input/output (I/O) ports. In hub, a signal introduced at the input of any port appears at the output of every port except the original incoming. A HUB works at the physical layer (layer 1) of the OSI model. Hubs are now largely obsolete, having been replaced by network switches except in very old installations or specialized applications.

Repeaters

- Regenerate the signal
- Provide more flexibility in network design

- Extend the distance over which a signal may travel down a cable
- Example → Ethernet HUB
- Connect together one or more Ethernet cable segments of any media type
- Works at Layer 1 of OSI model
- Forwards every frames; has no filtering capability

Bridge

- Join two LAN segments (A,B), constructing a larger LAN
- Filter traffic passing between the two LANs and may enforce a security policy separating different work groups located on each of the LANs.
- Bridges work at the Media Access Control Sub-layer of the OSI model
- A bridge stores the hardware addresses observed from frames received by each interface
- Uses this information to learn which frames need to be forwarded by the bridge.
- Each bridge has two connections (ports) and there is a table associated with each port.
 - A bridge observes each frame that arrives at a port
 - extracts the source address from the frame
 - and places that address in the port's routing table

Switch

- Typically connects individual computers
- A switch is essentially the same as a bridge though typically used to connect hosts, not LANs
- Richer management capability.
- Logically partition the traffic to travel only over the network segments on the path between the source and the destination (reduces the wastage of bandwidth)

Benefits/Advantages

- Improved security
- users are less able to tap-in into other user's data
- Better management
- control who receives what information (i.e. Virtual LANs)
- limit the impact of network problems
- Dedicated access
- Host has direct connection to the switch
- rather than a shared LAN connection

Cut-Through Switching

- Start transmitting as soon as it reads destination address.
- Transmit the head of the packet via the outgoing link while still receiving the tail via the incoming link

- Much faster
- Cannot detect corrupt packets
- Destination is responsible for detecting corrupt packets
- Can propagate the corrupt packets to the network

Store and Forward switching

- Read the whole packet before transmit
- Slower than the cut-through mode
- More accurate since corrupt packets can be detected using the FCS
- More suit to large LAN since they will not propagate error packets

Router

- Layer 3 Devices
- Network Layer in OSI and Internet layer in TCP/IP protocol stack
- Main Function
 - Routing of Packets (Based on Routing Table)
 - Path Selection(Best Path) to forward the packets
 - Internetwork Communications
- Only packets with known network addresses will be passed - hence reduce traffic
- Routers can listen to a network and identify its busiest part
- Will select the most cost effective path for transmitting packets
- Routing table is formed based on communications between routers using “Routing Protocols”
- Routing Protocols collect data about current network status and contribute to selection of the best path

Comparison of hub, switch and Router

	Hub/ Repeater	Bridge/ Switch	Router
Traffic isolation	no	yes	yes
Plug and Play	yes	yes	no
Efficient routing	no	no	yes
Cut through	yes	yes	no